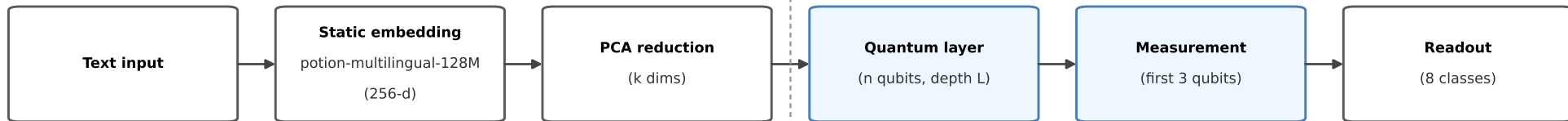


classical front end    quantum layer (simulated)



### Cross-validation protocol

Every circuit is evaluated twice:

- CUDA-Q (formal track; the reported numbers)
- an independent NumPy state-vector simulator

Agreement criterion:  $\max |\Delta P| < 1e-10$ .

Both tracks share one golden-vector test set,  
so a silent divergence cannot pass unnoticed.

